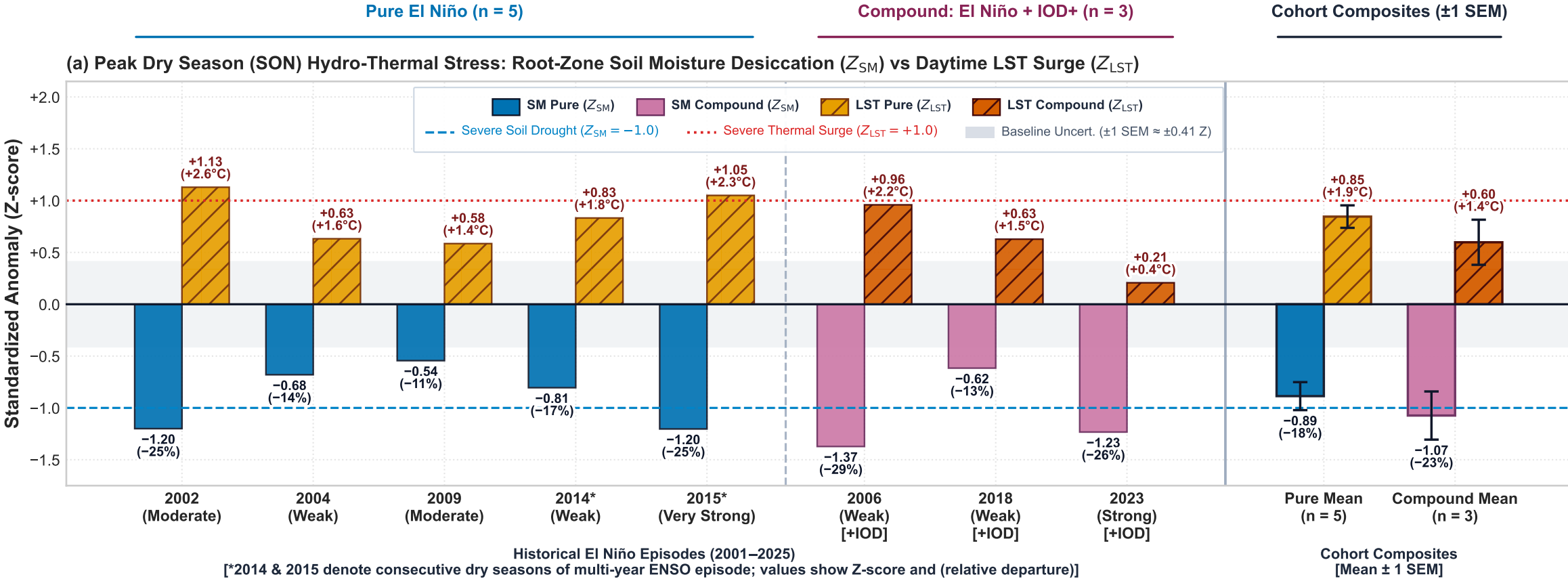
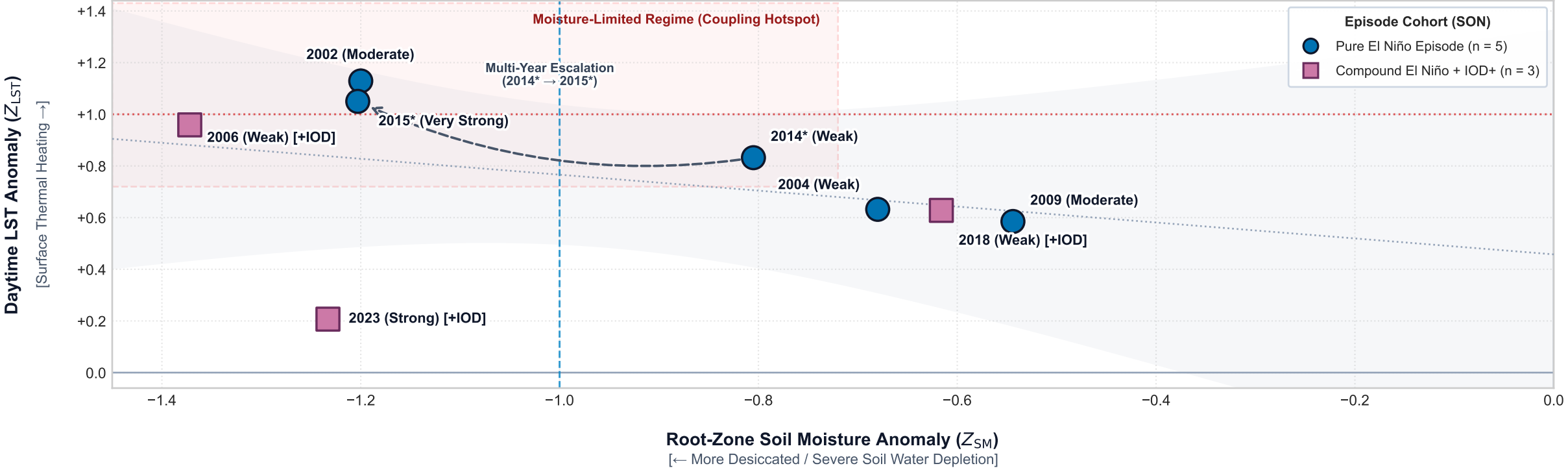


East Java Terrestrial Physical Response Across Historical El Niño Episodes (2001–2025)

ENSO-Neutral Baseline (n = 6; 2019 IOD+ unexcluded introduces slight conservative bias in Z-scores): Root-Zone SM = 0.237 m³/m³, LST = 33.3°C | Data: ERA5-Land (0–28 cm) & MODIS Terra MOD11A2 (1 km)



(b) Land-Atmosphere Coupling: Evaporative Cooling Breakdown across Historical Episodes (SON)



• Data Sources: ERA5-Land root-zone SM (0–28 cm) & MODIS Terra MOD11A2 1 km daytime LST.
• ONI Tiers: Weak (+0.5–0.9°C), Mod (+1.0–1.4), Strong (+1.5–1.9), V.Strong ($\geq +2.0^\circ\text{C}$); Compound (+ DMI $\geq +0.4^\circ\text{C}$).
• Episode Continuity: *2014 & 2015 = consecutive dry seasons of one ENSO episode ($n_{\text{eff}} < 8$).

• Neutral Baseline: 6 neutral years (2019 IOD+ unexcluded; baseline SEM = ± 0.41 Z).
• Hydro-Thermal Coupling: Evaporative shutdown ($\lambda E \rightarrow 0$) drives sensible heating surge ($H \uparrow \uparrow$).
• Statistical Rigor: Welch t (n=5 vs 3): SM t = 0.73 (p = 0.51, n.s.); LST t = 0.98 (p = 0.40, n.s.).